# Climbing Shoe Advisor App - Complete Project Roadmap

## 1. Project Vision

Build a climbing shoe recommendation platform that helps climbers identify the best shoe based on foot shape, climbing style, experience level, comfort preference, sizing, rubber characteristics, and intended usage (indoor/outdoor, bouldering/sport climbing).

## 2. Business Problem

Many climbers struggle to choose the right shoe. Online descriptions are often generic and sizing differs significantly between brands and models. The app should provide personalized recommendations similar to an experienced climbing-shop employee.

## 3. Target Users

Beginners buying their first shoe; intermediate climbers upgrading shoes; advanced climbers seeking performance models; climbing gyms; climbing retailers.

## 4. MVP Features

User questionnaire, shoe recommendation engine, shoe database, sizing guidance, explanation of recommendations, shoe comparison tool.

## 5. User Inputs

Foot width, foot volume, toe shape, street shoe size, climbing level, preferred climbing style, indoor/outdoor preference, comfort vs performance preference, softness/stiffness preference, budget.

## 6. Shoe Database Design

Create a database containing brand, model, softness score, stiffness score, aggressiveness, width, volume, rubber type, climbing discipline, skill level, sizing notes, pros and cons.

## 7. Recommendation Engine (Phase 1)

Use rule-based scoring. Assign points based on user preferences and shoe characteristics. Rank shoes by total score and display top recommendations.

## 8. Recommendation Engine (Phase 2)

Collect user feedback and build machine learning recommendations based on successful shoe selections and user satisfaction.

## 9. UI/UX Design

Home page, questionnaire wizard, recommendation results page, shoe details page, comparison page, user profile.

## 10. Technology Stack

Frontend: React + Tailwind CSS. Backend: Python FastAPI. Database: PostgreSQL or Supabase. Authentication: Clerk/Auth0/Supabase Auth. Hosting: Vercel + Railway/Render.

# 11. Database Schema

Tables: Users Shoes Brands Recommendations Reviews Sizing_Guides User_Feedback

# 12. Data Collection Strategy

Begin with 20-50 popular shoes from La Sportiva, Scarpa, Tenaya, Ocun, Red Chili, Evolv, Butora, Unparallel, So iLL and Mad Rock. Expand gradually.

# 13. Development Phases

Phase 1: Spreadsheet prototype. Phase 2: Rule-based web application. Phase 3: Database integration. Phase 4: User accounts. Phase 5: Feedback collection. Phase 6: Machine learning recommendations.

# 14. Advanced Features

Inventory integration for climbing stores, QR-code shoe finder, shoe similarity search, resole recommendations, AI assistant, multilingual support, retailer dashboard.

# 15. Portfolio Value

Demonstrates product thinking, database design, business analysis, recommendation systems, frontend development, backend development, SQL, analytics, and domain expertise in climbing equipment.

# 16. Suggested Timeline

Month 1: Database and research. Month 2: MVP questionnaire and recommendation logic. Month 3: Frontend implementation. Month 4: Backend and database integration. Month 5: Testing and deployment. Month 6: Feedback and advanced features.

# 17. Long-Term Vision

Become the 'Bike Fitting Tool' equivalent for climbers by helping users find their ideal shoe fit and performance characteristics while supporting retailers and climbing gyms.